

SHUFAN SHAHI

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EDUCATION

Islamic University of Technology	2022–2026
B.Sc. in Computer Science and Engineering	CGPA: 3.55
Coursework: Algorithms, Machine Learning, Deep Learning, Databases, and Networks.	
Chuadanga Govt. College	2021
Higher Secondary Certificate (HSC)	GPA: 5.00/5.00

EXPERIENCE

Student Researcher	2026–Present
<i>Elite Research Lab</i>	
Support research projects through literature reviews, experiments, and analysis.	
Researcher	2025–2026
<i>IUT Computer Vision Lab</i>	
Researched multimodal AI methods to improve integration and mitigate modality imbalance.	
Senior Executive	2025–2026
<i>Project Aero</i>	
Contributed to drone R&D by applying AI-based object tracking for international UAV competitions.	

PUBLICATIONS AND CONFERENCES

EthMuSAM : Mutual Scoring With SAM For Zero Shot Anomaly Segmentation[Preprint]	2026
- Extracted patch-level representations from images using pretrained DINOv2 encoders. - Localized anomalous regions by comparing test-image embeddings with representations from normal reference images.	
Prototype-Guided Adaptive Feature Learning for Multimodal Emotion Recognition in Conversation (MERC)[Preprint]	2026
- Developed graph-compatible adaptive feature weighting and modality-specific prototype supervision to learn class-discriminative representations before cross-modal reasoning. - Improved multimodal emotion-recognition performance across multiple benchmark datasets through adaptive feature processing and prototype-guided learning.	
Rethinking Visual Features in Graph Based Multimodal Emotion Recognition in Conversations (MERC): From Low Level Pixels to Semantic Representations[Preprint]	2026
- Developed a semantic visual-feature pipeline that converts utterance-level videos into temporally structured descriptions of facial expressions, gaze, body language, and emotional transitions. - Encoded the descriptions into dense representations for graph-based multimodal reasoning, improving cross-modal alignment while providing a viable alternative to noisy handcrafted visual features.	

PROJECTS

DocuMind - RAG Document Q&A Assistant | *Python, LangChain, LlamaIndex, ChromaDB, Ollama*  [GitHub](#)

- Built a local RAG system for documents with cited answers and hallucination guardrails.
- Compared 4 chunking methods and 2 embedding models; added hybrid BM25 and reranking.

Drone Vision - Aerial Tracking | *PyTorch, YOLOv11X, ByteTrack*  [GitHub](#)

- Fine-tuned YOLOv11X on VisDrone and integrated ByteTrack for human counting and multi-object tracking in aerial footage.

BD Traffic Monitoring System - Traffic Analytics | *PyTorch, YOLO11L, Flask, Docker*  [GitHub](#)  [Live](#)

- Fine-tuned YOLO11L on Bangladesh traffic data to detect local vehicle classes and deployed the inference application with Flask and Docker.

Biztrack - Business Analytics | *Next.js, Node.js, PostgreSQL*  [GitHub](#)  [Demo](#)

- Built AI modules for real-time financial tracking, risk alerts, customer segmentation, and forecasting.

TECHNICAL SKILLS

Programming Languages: Python, C++, Java

Data and Machine Learning: PyTorch, TensorFlow, scikit-learn, NumPy, Pandas

Databases: MySQL, Firebase, PostgreSQL

Web Development: React, Tailwind CSS, Next.js, Spring Boot

Mobile Development: React Native, Flutter

Visualization: Matplotlib, Seaborn

Development Tools: Git, Linux

ACHIEVEMENTS, AWARDS AND SCHOLARSHIPS

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| • 5th Place, VAND4 International Competition | 2026 |
| Zero-Shot Industrial Track; built a zero-shot computer vision pipeline for detecting anomalies in industrial images using foundation models. | |
| • First Runner-up, IUT Coderush Hackathon | 2025 |
| Built an application under tight deadlines in a fast-paced team. | |
| • 7th Place, BAIC Datathon | 2025 |
| Analyzed datasets using preprocessing, visualization, and modeling. | |
| • IUT Partial Scholarship | 2022 |
| Secured a partial scholarship covering all four years of B.Sc. studies. | |
| • Runner-up, State-Level Physics Olympiad | 2018 |

REFERENCE

Shahriar Ivan

Assistant Professor, Department of CSE, IUT

Specialization: Machine Learning, Deep Learning, Computer Vision

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